



SMART
INDUSTRY

SMART CONVEYOR AUTOMATION SYSTEM

MINI PROJECT REPORT

1. Introduction

Nowadays, factories are increasingly relying on smart control systems instead of traditional methods that require a lot of human intervention. One of the most important of these systems is the PLC system, which is used to control production lines with high accuracy and speed.

In this project, a simple model was designed to simulate a production line that includes a conveyor belt that automatically transports products, using sensors to detect product movement and controlling the motor operation automatically.

The main idea is to reduce manual intervention and increase the speed and accuracy of handling products during movement inside the production line.

2. Objective of the Project 🎯

This project is designed to:

- Understand the concept of automation control in factories
- Learn the role of PLC in managing production lines
- Use sensors to track products
- Control motor movement automatically
- Reduce errors caused by manual intervention
- Simulate a simple real industrial system

3. System Idea

The system consists of a simple conveyor belt on which a product moves, along with two sensors:

- The first sensor detects when a product enters the conveyor
- The second sensor detects when the product exits or reaches a specific point

Once the product passes in front of the sensors, the system records this movement and keeps the conveyor running automatically.

At the same time, there is a start button and a stop button for full system control.

4. Components Used🔧

The system consists of:

- **PLC Controller:** The main control unit that makes decisions
- **IR Sensors:** Detect product movement without contact
- **DC Motor:** Responsible for conveyor belt movement
- **Conveyor Belt:** Transports the products
- **Push Buttons:** Start and stop control buttons
- **LED Indicators:** Show system status (Running / Stopped)

5. How the System Works 🔄

The operation follows a simple sequence:

- When the Start button is pressed, the system begins operating
- The motor starts moving the conveyor belt
- When a product passes the first sensor, its entry is recorded
- When it reaches the second sensor, completion is recorded
- The conveyor keeps running as long as the system is ON
- When the Stop button is pressed, the entire system stops immediately

The whole idea is based on a simple sequence of signals between sensors and the PLC.

6. System Layout ¶ (Wiring Concept)

The system is connected as follows:

- Sensor 1 → PLC Input
- Sensor 2 → PLC Input
- Start/Stop Buttons → Inputs
- Motor → PLC Output
- LEDs → Outputs

Each input and output controls a specific part of the system operation.

7. Control Logic (Simplified)

- If Start = ON → Motor starts running
- If Sensor 1 = ON → Product Entered is recorded

- If Sensor 2 = ON → Product Processed is recorded
- If Stop = ON → Motor stops

The entire system is based on a simple “If / Then” logic, but in an industrial form.

8. Where This Is Used 🏭

This system is similar to what is used in:

- Packaging factories
- Food production lines
- Pharmaceutical factories
- Sorting systems
- Assembly lines

★ 9. Advantages

- Automatic operation without continuous human intervention
- Faster production speed
- Reduced human errors
- Easy control
- Suitable for both small and large factories

10. Limitations⚠️

- Requires initial cost

- Needs regular maintenance
- Any sensor failure affects the system
- Requires basic PLC knowledge

11. Future Improvements

The system can be improved by:

- Adding a digital product counter
- Connecting the system to a display screen
- Using a camera instead of sensors
- Controlling via mobile phone
- Adding a smart product sorting system

12. Conclusion

In conclusion, this project demonstrates a simple idea of how modern factories operate automatically using PLC systems and sensors.

Although the model is simple, it reflects a core concept used in large industrial systems and helps in understanding the basics of automation control in an easy and practical way.

INPUT DEVICES



IR SENSOR 1
(Product Entry)



IR SENSOR 2
(Product Exit)



**START
BUTTON**



**STOP
BUTTON**

CONTROLLER



PLC
(Controller)

OUTPUT DEVICES



DC MOTOR
(Conveyor Drive)



GREEN LED
(System ON)



RED LED
(System OFF)



POWER SUPPLY
(24V DC)